

InaTechShips (~28k images)

Close-range ship photos from shipspotting.com

SAM ViT-B segmentation

Bounding box prompt → RGBA mask

23,828 ship crops

Quality-filtered RGBA (85.7% yield)

CITRA-3D-Real (2,081 images)

Operational naval surveillance scenes

Extract bounding boxes

Position + scale + count per image

7,003 anchor positions

Real vessel locations × 13 variations

In-place substitution

Resize each crop to bbox dimensions → alpha blend onto scene

Labels = original annotations (100% bbox accuracy)

*×13 variations
per background*

Synthetic dataset (27,796 images)

93,480 objects · 3.36 obj/img · split 60/20/20